Accounting and Measurement

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October 30, 2026

Outline of Talk

1. Measuring the "state" of the economy
   - What measurement does.
   - Why do we care?
2. GDP
   - Expenditure Approach
   - Income Approach
   - Production Approach
   - What's in GDP and what's not?
3. Changes Over Time
4. Comparative Economic Performance
5. Summary
6. Stickers for the Brain

Measuring the State of the Economy

Accounting and Measurement



- Accounting is a very important discipline.
  - It helps keep track of the **stock** of resources in a given entity (i.e. home; business; government; overall economy).
    - ★ A **stock variable** measures the amount of a resource at a given tim period; a.k.a. a **state variable.**
    - ★ Examples: total output; capital stock.

Measuring the State of the Economy

Accounting and Measurement

- Accounting provides accurate snapshots of how the **stock** is formed ...
- i.e. what are the **flow variables** contributing to it, and how much.
  - A **flow variable** measures amounts of resources per unit of time.
    - ★ Flows culminate in stocks.
    - ★ Example: consumption, investment, imports and exports per quarter.



Accountants: the Guide Dogs of the economy.

Measuring the State of the Economy

Why do we care?

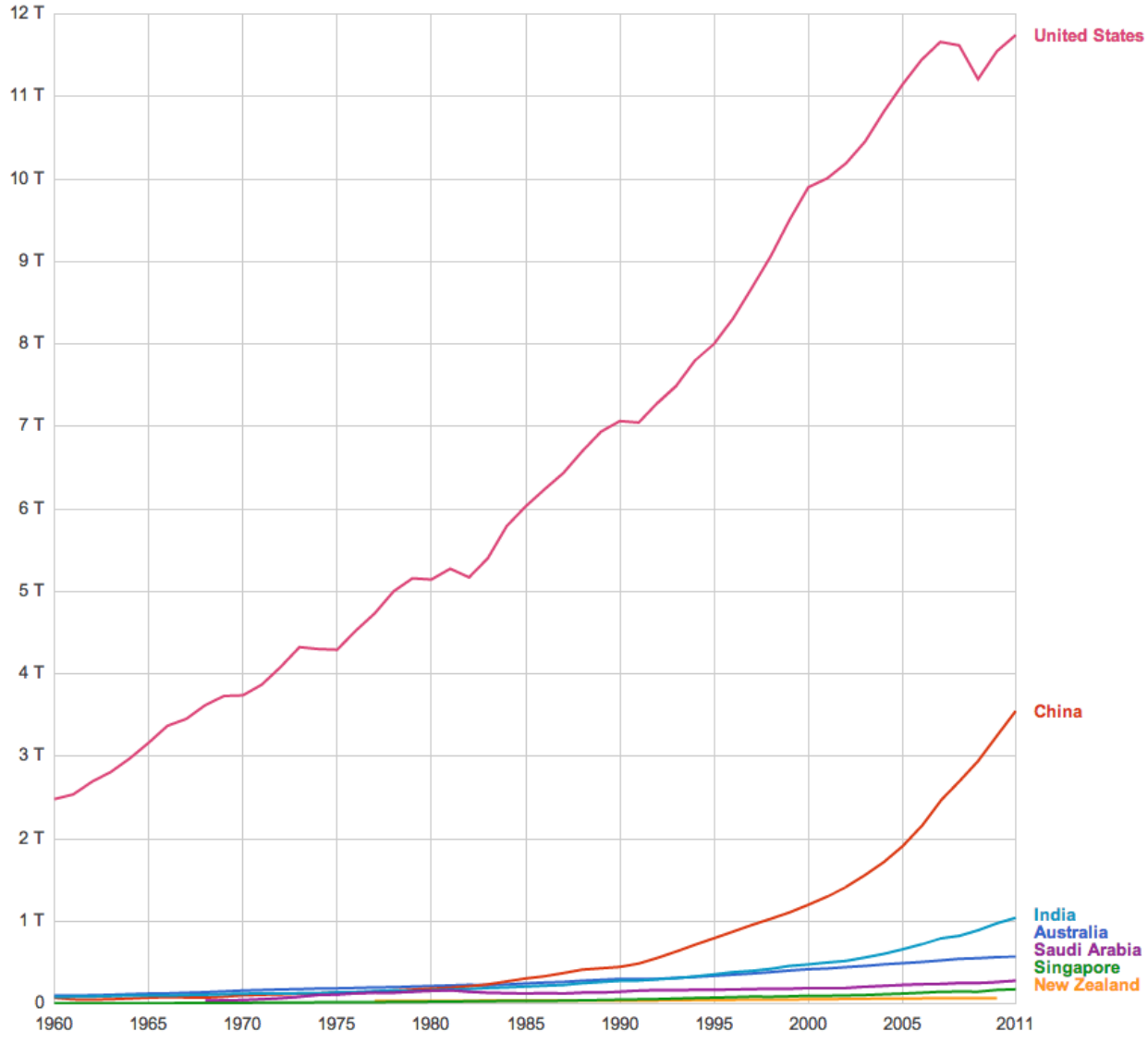
- We would like to understand our economic history: the economy's historical behavior or outcomes.
- Knowing the state of the economy over time allows us to form a v about economic growth and allocation of resources over time.
- A good measurement of economic activity over time helps shape better models.
- Better models help inform understanding of the data and better policy designs.

National Income

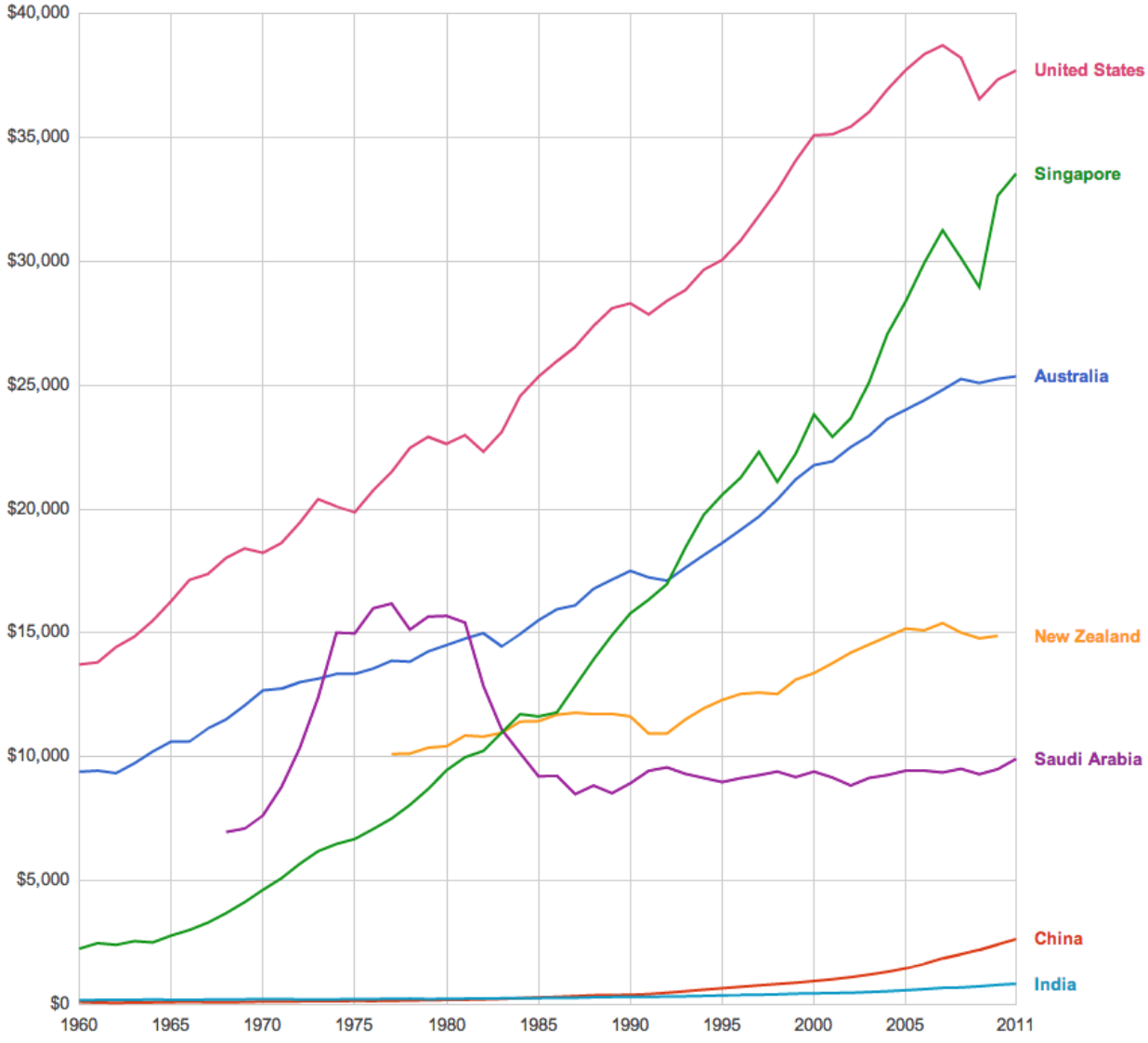
Why do we care? An economic development Example

Let's bring home these points a bit more ...

- It's more fun looking at pictures.
- Let's look at some so-called "long run" data as an example.
  - These are annual country observations over many, many years.
  - The data measures each country's
    - "real GDP" (i.e. a single total measure of economic production/income in terms of *quantities* of goods and services); and
    - "real GDP per capita" (i.e. real GDP divided by total population size). Often called "average living standard" or "living standard", in popular discourse.



Real GDP as total of expenditure flows (in constant 2000 US dollars).



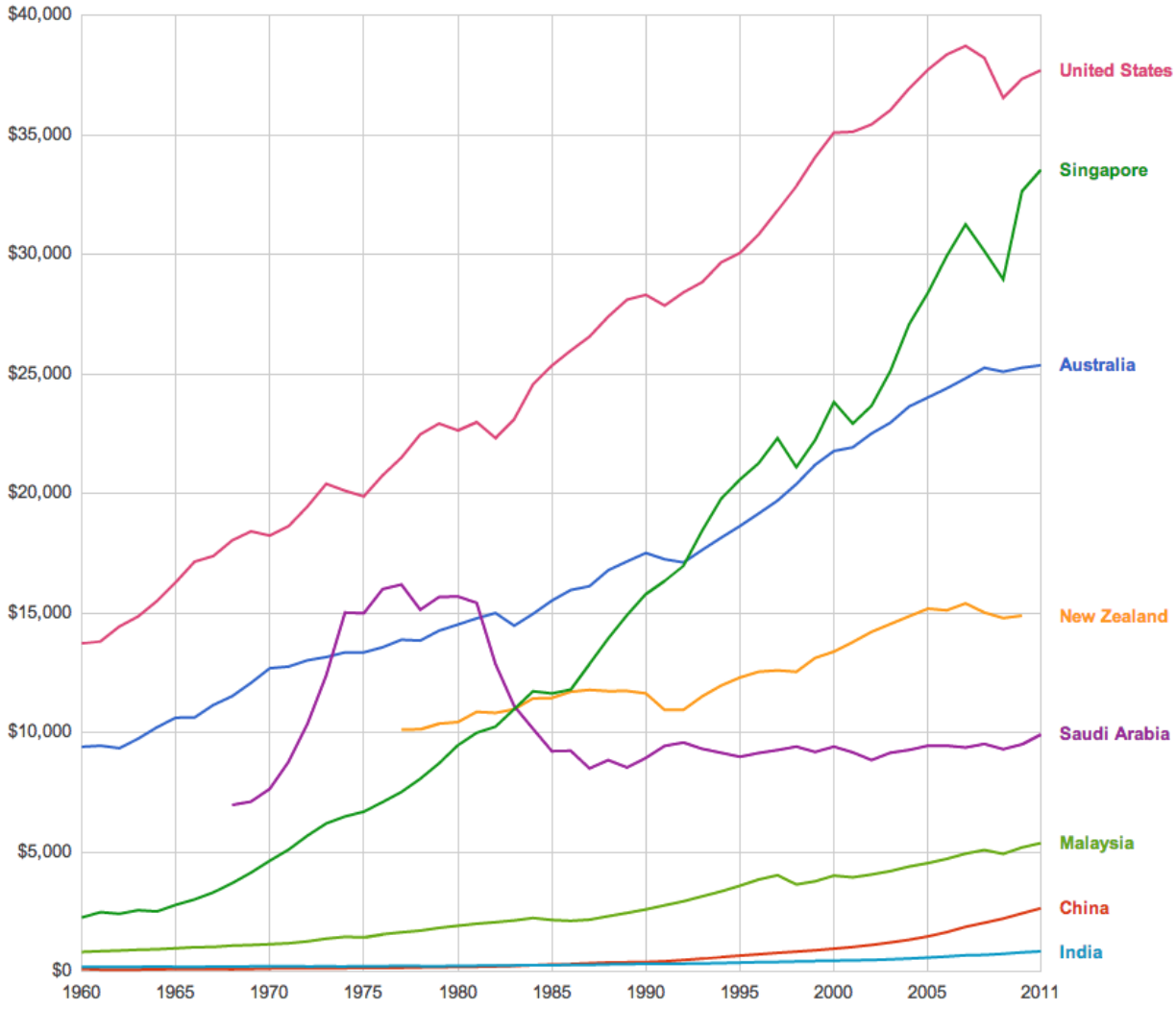
"Average Living Standard": Real GDP per person as total of expenditure flows (in constant 2000 US dollars).

National Income

Why do we care? GDP as a measure of economic state

Here's an interesting observation ...

Suppose I add data on another country ... say Malaysia.



"Average Living Standard": GDP per person as total of expenditure flows in constant 2000 US dollars.

National Income

Why do we care? GDP as a measure of economic state

We just had an interesting observation ...

I just added "living standard" data on another country ... Malaysia.

Comment on this chart.

Hint: Look at its closest geographical neighbor ... Imagine the view forward from 1963-1965.



Ranking of "Average Living Standard" y:

$y_{Singapore} > y_{Australia} > y_{Malaysia} > y_{China}$.

Measuring GDP

- We just discussed an example relating to economic development
- Usefulness and importance of knowing what each country's econ activity is.
- Next, we look at the break down on what is this convenient mea of economic activity called GDP.
  - Composition. Three ways to measure composition:
    - ★ income,
    - ★ expenditure, and
    - ★ production.

Measuring GDP

- A conventional measure for the state of an economy is **gross domestic product (GDP).**
- GDP is the *market value* of *final goods* and *services* measured by the end of a given period of time.
  - Example: Consider a one-year period.
    - ★ Account for all the cars, massages, concerts, books, paper-clips, etc, produced.
    - ★ These goods are services must have a market price. (So your bootleg DVD does not count.)
    - ★ Add their value up for that year.
    - ★ You get a measure of GDP.

Measuring GDP

There are three ways to skin a cat

- There is more than one way to measure GDP.
  1. **Production measure:** counts the number of goods produced in the economy.
  2. **Expenditure measure:** counts the total purchases in the economy.
  3. **Income measure:** counts all the income earned in the economy.
- All three approaches give identical measures of GDP.

Measuring the State of GDP

Things must add up!

Let's put on our accountant hats ...

The national income accounting identity says:

$$Y = C + I + G + NX$$

- Y is income or total output produced
- C is total consumption
- I is total investment
- G is government expenditure
- NX is exports X minus imports M

Measuring the State of GDP

Things must add up!

National income accounting identity:

$$Y = C + I + G + NX$$

- LHS: can be measured in two ways:
  - add up all market value of income earned (GDP income approach); or
  - add up all market value of goods and services produced (GDP production measure).
- RHS: Total expenditure, measured by:
  - adding up all spending by domestic and foreign demands, and
  - less domestic demand for foreign goods and services
  - (GDP Expenditure approach).

Measuring the State of GDP

Things must add up!

Example

Wilbur and Fred own the Bananarama Republic (BR), a farm business. They have workers who pick banana crops. Think of BR as an economy. It's balance sheet is its national account. The harvested bananas are all sold at the local market, where the workers go shopping as well.



Measuring the State of GDP

Things must add up!

In the example:

- The **income approach** would measure the sum of
  - all the wage bill paid to the workers; and
  - the profits* to Wilbur and Fred.
  - ***Economic profits**: super-normal returns associated with prices that exceed what would have been under perfect competition.

Measuring the State of GDP

Things must add up!

In the example:

- The **production approach** would measure the sum of
  - all the bananas picked by each worker.
- The **expenditure approach** would measure the value of
  - all bananas sold in the market.
  - if Wilbur, Fred and their workers were considered citizens of BR, we can count bananas bought by:
    - ★ themselves are domestic expenditure; and
    - ★ others outside of BR as exports.
  - as a variation on the example, if BR had to buy a truck from the market outside, then it counts as BR's import.

Expenditure Approach to GDP

The **expenditure measure** of GDP counts the total purchases of goods and services in the economy.

Let's look at the RHS of the national income account in more detail:

$$Y = C + I + G + NX$$

Now, more precisely ...

- C is total consumption by households on goods and services
- I is total investment expenditure for the purposes of adding to production
- G is government purchases on goods and services*
- $NX = X - M$, is net exports: exports X minus imports M

* What about government spending on transfers, e.g.:
  - social security or pensions,
  - medicare, or
  - unemployment insurance?

... or on paying interest on government debt?

Expenditure Approach to GDP

	Total (billions of dollars)	Share of GDP (percent)	Per person (dollars)
Gross domestic product	12,460	100.0	42,000
Consumption	8,740	70.1	29,470
Motor vehicles and parts	450	3.6	1,510
Food	1,200	9.6	4,050
Housing	1,300	10.4	4,400
Medical care	1,490	12.0	5,030
Investment	2,060	16.5	6,940
Structures (nonresidential)	340	2.7	1,140
Equipment and software	930	7.5	3,130
Residential	770	6.2	2,600
Government purchases	2,370	19.0	8,000
National defense	590	4.7	1,990
Net exports	−720	−5.8	−2,420
Exports	1,300	10.4	4,390
Imports	2,020	16.2	6,810

Source: U.S. Department of Commerce, Bureau of Economic Analysis, www.bea.gov.

TABLE 2.1 The Expenditure Approach to U.S. GDP in 2005

Macroeconomics, Economic Crisis Update
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Consider arbitrary year 2008 ...

	($bn)	%				
Country	*GDP*	*C*	*I*	*G*	*X*	$-(-M)$
US	14,265	71	14	20	13	20
Japan	4294	55	25	18	18	16
Germany	2928	55	18	18	47	41

What can you say about the expenditure components making on total GDP expenditure?

Income Approach to GDP

- The **income approach** measures the sum of all income earned in the economy.
- Incomes appear as:
  - wages and salaries
  - worker benefits (e.g. health, dental)
  - economic profits ("net operating surplus of businesses")
  - depreciation of "capital" *

Income Approach to GDP

* Why include depreciation of capital as income?
  - Capital is the total of inputs into production other than labor that are not used up in the production process.
    - Firms increase capital stock through investment flows.
  - Depreciation is the deterioration of the capital stock due to wear and tear.
  - Implicitly some of the income earned must be used to pay for depreciation.
  - So we need to "add it back in" in the accounting exercise.

Income Approach to GDP

	Total (billions of dollars)	Share of GDP (percent)	Per person (dollars)
Gross domestic product	12,460	100.0	42,000
Compensation of employees	7,040	56.5	23,730
Wages and salaries	5,670	45.5	19,110
Benefits	1,370	11.0	4,620
Indirect business taxes	870	7.0	2,930
Net operating surplus of business	2,880	23.1	9,710
Depreciation of fixed capital	1,600	12.8	5,390

Source: U.S. Department of Commerce.

TABLE 2.2 The Income Approach to U.S. GDP in 2005

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Income Approach to GDP

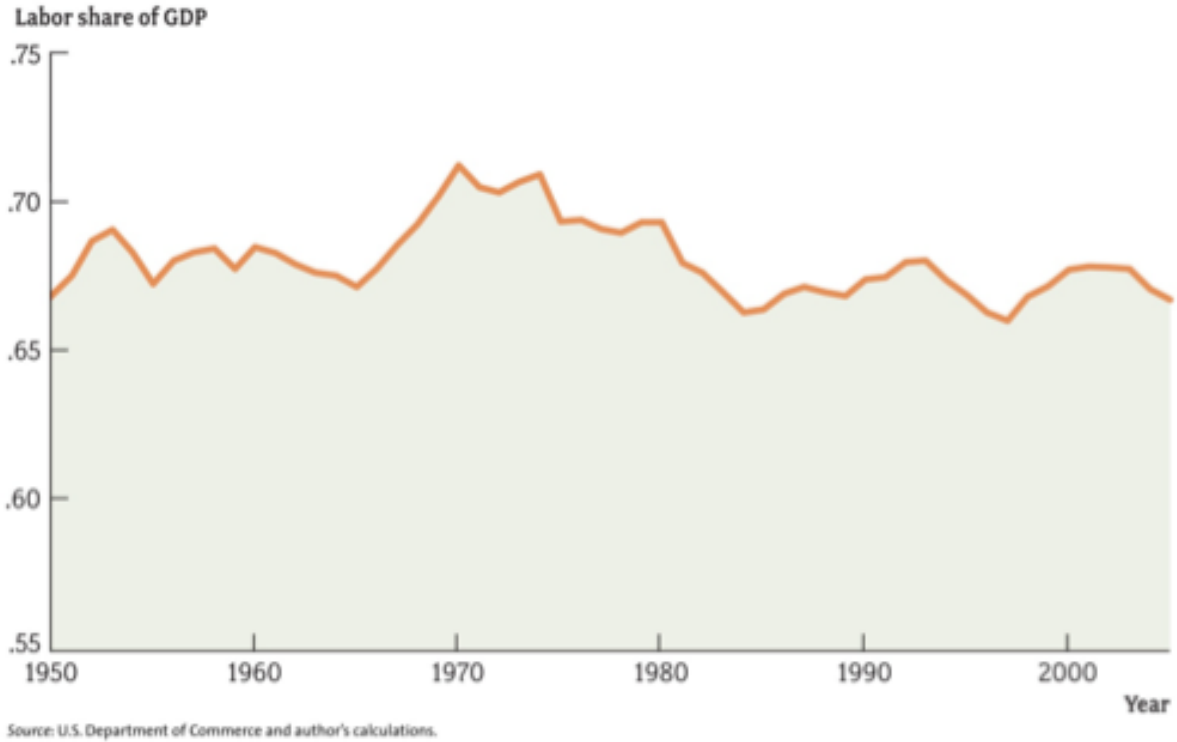
Another way to visualize national income is:

- Classify all income figures as either labor or capital.
- this is another artificial construct.

If we imagine it so ... the data produces the following empirical regularity for most countries:

- The total share of GDP in terms of labor income is approximately two-thirds (long-run average); and
- the total share of GDP to capital is approximately one-third.
- Labor's share of GDP has remained around a constant average over time.

Income Approach to GDP



Source: U.S. Department of Commerce and author's calculations.

FIGURE 2.3 Labor's Share of GDP, 1950–2005

Macroeconomics, Economic Crisis Update

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Production Approach to GDP

- "Value of goods and services produced in an economy over a given period" ...
- Not precise enough.
- Principle: no double counting of value of production.
- Measurement Concept: **Value added**
  - The amount each producer in the economy contributes to GDP.
  - The revenue generated by each producer minus the value of intermediate products.
- So more precisely, GDP (production) measures the total of value added.
- Equivalently, only **new production** of goods and services count towards this approach of measuring GDP.

Production Approach to GDP

- Only goods and services that are transacted through markets are included in GDP.
- GDP does not include:
  - Government transfer payments to individuals.
  - Social Security, Medicare, unemployment insurance.
  - A measure of the health of a nation's people.
  - Changes in environmental resources.

Changes Over Time

Real vs. Nominal

- Recall previous motivating example: comparison of real GDP or capita real GDP across nations, over time.
- Problem with measuring "real GDP". Aggregate quantity not di measurable.
- How then do we solve this problem?
  - First, we can measure **nominal GDP**, i.e. economic activity in "c market value" or money/nominal terms.
  - Second, we deal with a problem of **inflation** and changes of mon value over time, to construct **price indexes**.
  - Third, we construct measures of **real GDP**.
- Here I will expand on the textbook HGLO. Read the WATTLE handout too.

Changes Over Time

Real vs. Nominal

- Two measurement concepts:
  - **Nominal GDP:** A measure of GDP when **prices** and **quantities** have not been separated. So components of the identity

$$Y = C + I + G + X - M$$

    are measured in current money values.
  - **Real GDP**: Actual **quantity** of goods and services.

Changes Over Time

Real vs. Nominal: observables?

- So we have the relation:

$$Y_{nominal} = P \times Y_{real}$$

  where
  - $Y_{nominal}$ is Nominal GDP
  - P is aggregate price level
  - Y_{real} is Real GDP
- But where does the aggregate price level P come from?
- Do we observe Y_{real} ?
- We do observe $Y_{nominal}$, or at least we can account for it readily from the three approaches studied earlier.

Changes Over Time

Constructing Nominal GDP

Example

An economy with two types of goods/services, haircuts (H) and jackfruits (J). Then ...

$$Y_{nominal} = P_H \times Q_H + P_J \times Q_J$$

where P_i is price level of good i, and, Q_i is quantity of good i; and i is either H or J.

Changes Over Time

Constructing GDP

- For concreteness, we defined a money/nominal price level in "dollars". (Substitute with "baht", "yuan", etc. as needed.)
- To compute GDP across time, we must use one year's price.
  - Real GDP will be measured in a certain year's dollars.
  - Nominal GDP is measured in current dollars.
- If the quantity of goods and services produced does not change, but prices do change, then:
  - Nominal GDP will change, but ...
  - Real GDP will not change.

Changes Over Time

Example: when real GDP does not change

Example

Consider our two-good (H and J) economy again.

	Period 1	Period 2
Q_H	100	100
Q_J	200	200
P_H	20	22
P_J	4	6

Changes Over Time

Example: when real GDP does not change

- Nominal GDP has increased from $2800 in Period 1 to $3400 in Period 2.
- Period 1 Real GDP in Period 1 dollars is $2800.
  - (Note same as Period 1 Nominal GDP!)
- Period 1 Real GDP in Period 2 dollars in $3400.
  - (Same as Period 2 Real GDP in Period 2 dollars; same as Period 2 Nominal GDP!)
- But we know quantities in both good/service have not changed over the two periods!
- Here all the changes in (Nominal) GDP is attributed to changes in the price levels over time.

Changes Over Time

Remarks

- Confusingly, although Real GDP is said to measure actual "quantitites" it is still measure in "dollars", just like Nominal GDP.
- Why?
  - Trivial in a one-good economy. Then real GDP is literally the quantity of single good.
  - In a multiple goods/services economy, we can't literally have a single quantity measure for haircuts, jackfruits, cars, etc.
  - Put simply, we cannot directly compare apples and oranges.
  - To solve this problem we need an index of prices — a way to aggregate the price levels of different goods and services.
  - More on this later ...
- We deal with another example first ...

Changes Over Time

Example: when real GDP does not change

Example

Consider our two-good (H and J) economy again, with the following new data in Period 3 ...

	Period 1	Period 2	**Period 3**
Q_H	100	100	**110**
Q_J	200	200	**240**
P_H	20	22	**30**
P_J	4	6	**6**

Changes Over Time

Example: when real GDP does not change

- Compared to Period 2, the economy is producing/consuming 10 more haircuts and 40 more jackfruits in Period 3.
- Intuitively, Real GDP should now change, and be higher in Period 3.
- Show that:
  - Period 3 Real GDP is \$3,860 in Period 2 prices.
  - Period 2 Real GDP is \$3,400 in Period 2 prices.
  - So Real GDP, measured in Period 2 prices, has increased over Period 2 to 3, by about 13.53%.
- Now do the same using Period 3 prices as the basis. How much has Real GDP increased over Period 2 to 3? Is it about 12.86%?

Try this as an exercise ...



- The two methods give the same qualitative answer, but slightly different quantitative percentage growth in Real GDP!
- This example shows that what "base year" we use as constant-price year matters for real GDP values over time.

Changes Over Time

Quantity Indexes

- Recall last example.
- Calculating real GDP changes over time:
  - The **Laspeyres index**: Calculates changes in real GDP using the initial prices.
  - The **Paasche index**: Calculates changes in real GDP using the final year prices.
- Over long-time intervals the two indexes can result in substantial differences.

Changes Over Time

Quantity Indexes

- The **Fisher index** (**chain weighting**) is the preferred approach to calculating real GDP.
- Property of this index:
  - a (weighted) average of the Laspeyres and Paasche index.
- Preferred because:
  - Takes into account **obsolescence** and **new products**:
    - ★ new goods/services are invented,
    - ★ while others become obsolete,
    - ★ making early or recent prices inaccurate.
  - Can be applied on a year-by-year basis if we compute real GDP each year.



The secretarial typing pool of the 1950s: Obsolete service.



The personal computer and office productivity software: New products.

Changes Over Time

Example: Chain-weighted Real GDP

Example

Consider our two-good (H and J) economy again, as in last example ...

	Period 1	Period 2	**Period 3**
Q_H	100	100	**110**
Q_J	200	200	**240**
P_H	20	22	**30**
P_J	4	6	**6**

Changes Over Time

Example: Chain-weighted Real GDP

- We showed previously that:
  - Real GDP, measured in **Period 2** prices, has increased over Period 2 to 3, by about 13.53%. (**Laspeyres Indexing)**
  - Real GDP, measured in **Period 3** prices, has increased over Period 2 to 3, by about 12.86%. (**Paasche Indexing)**

Changes Over Time

Example: Chain-weighted Real GDP

- To construct a Fisher (Chain-weighted) Real GDP Index between Period 2 and 3:
  1. Average* the Real GDP growth rates based on Laspeyres and Paasche Indexes: $1/2 \times (13.53\% + 12.86\%) = 13.19\%$
     - ★ i.e. chain-weighted index of GDP in Period 3 is 13.19% higher than itself in Period 2.
  2. Calculate Real GDP measured in **Period 3** prices (= Nominal GDP in Period 3!). This is $\$4,740$.
  3. Back out chain-weighted measure of Real GDP in Period 2 as the unknown variable:

$$Y_2 = \frac{\$4,740}{1 + 0.1319} \approx \$4,188.$$

*Note: in Step 1, we took a simple/arithmetic average of the growth rates. More precisely, the Fisher index is computed by taking a "geometric average" instead.

Changes Over Time

Example: Chain-weighted Real GDP (Remarks)

- Inductively (backwards), we can apply the same idea for more than two periods of interest. (See text and tutorial.)
- Note we calculated the chain-weighted Real GDP for Period 2 as:

$$Y_2 = \frac{\$4,740}{1+0.1319} \approx \$4,188.$$

- The Laspeyres-index Real GDP for Period 2 is $\$3,400$.
- The Paasche-index Raal GDP for Period 2 is $\$4,200$.
- So we have verified the Property of the chain-weighted index Real GDP:
  - it is a (weighted) average of the last two measures.
  - so it has to be a number between (and possibly equal to one of) the two measures.

Changes Over Time

Price indexes and inflation

- Recall Nominal GDP is defined as

$$Y_{nominal} = P \times Y_{real}$$

- The GDP deflator is the price level P that satisfies the equation.
- We could compute this formula for two different periods to calculate a price change.
- We could also use the following math trick (approximation):
  - Percentage change in $Y_{nominal}$ approximately equals ...
  - Percentage change in P plus Percentage change in Y_{real}.
- Percentage change in P is called the **inflation rate of the GDP deflator**.

Changes Over Time

Using Chain-weighted data

- Main reason for using chain-weighted data:
  - Prices of computers rapidly changing in 1990s.
- Main disadvantage:
  - The sum of real C, I, G, NX will not equal real chain-weighted GDP (i.e. Y)
  - because the prices used in constructing the components are different.
- General rule to follow:
  - For particular components of GDP, we look at the ratio of nominal variables.
  - When you want real rates of economic growth, use the chain-weighted real measures.

Comparative Economic Performance

- We like to have orderings defined for comparable things.
- This allows us to assign rankings to them.
- Likewise, economists are interested in comparing economic out
  between countries, between states and etc.
- This allows us to make conclusions regarding issues like which
  has a higher living standard, and how far apart countries are in
  of economic development.

Comparative Economic Performance

- The exchange rate: Price at which different currencies are traded.
- Cross-country GDP figures must be expressed in a common currency by first adjusting it by the exchange rate. Why?
  - Price levels change over time, and, for each country.
  - Need to make two country's GDP comparable.
  - So adjust all to a common price series.
- This value of nominal GDP must be multiplied by the ratio of prices in the countries.

Comparative Economic Performance

Example

China and US. Suppose (nominal) exchange rate is \$1 exchanges for 7.6 yuan, $E = \$1/7.6 yuan$.

Step 1: Suppose an identical unit bundle of goods in China cost P_{China}, and in US costs, P_{US}.

Step 2: Let nominal GDP of China be GDP_{China}. In US dollars this is

$$GDP_{China}^{dollars} = GDP_{China} \times E.$$

Step 3: Then Real GDP of China in US Prices:

$$RGDP_{China}^{P_{US}} = \frac{P_{US}}{P_{China}} \times GDP_{China}^{dollars}.$$

Comparative Economic Performance

- Comparison of countries:
  - In general, rich countries tend to have higher price levels than poor countries.
  - This is mainly because poor countries have lower wages.

Summary

- National income accounting provides systematic measures o aggregate economic activity.
- Gross domestic product (GDP) is the key overall measure of activity in an economy.
- Can be viewed as total expenditure, total income, or total p in an economy.
- The expenditure approach to GDP makes use of a fundame national income identity: $Y = C + I + G + NX$
  - which says that total spending is the sum of spending on co investment, government purchases, and net exports.

Summary

- The income approach shows that labor's share of GDP is relatively stable over time at about two-thirds.
- In the production approach, it is only the value of final production that counts.
  - Equivalently, GDP is the sum of value added at each stage of production.

Summary

- Nominal GDP refers to the value of GDP measured in current prices in a given year.
- Real GDP involves computing GDP in two different years using the same set of prices.
  - Think about real GDP as being adjusted for inflation.
- Changes in real GDP therefore reflect changes in actual production rather than changes in prices.
- Chain weighting allows us to compare changes in real GDP over time by gradually updating prices.
- By linking the chain of comparisons in this way, we construct a more accurate measure of real GDP.

Summary

- International comparisons of GDP involve two conversions.
  - First, we need exchange rates to convert the measures into a common currency.
  - Second, just as we need to use common prices to measure real GDP over time, we also need to use common prices to compare real GDP across countries.

Measurement helps answer important questions



We will be asking: **Why** do some nations grow and become wealthier? ...

Measurement helps answer important questions



... and **why do** some others lag behind and remain in poverty?

More Scout Badges

... for your journey so far.

The following may be a useful list of keywords to help you recall concep discussed in this talk. ***Caution:*** *Apply sparingly.*

Key words:

- Accounting
- Stocks/states and flows
- Measurement, economic outcomes
- Accurate data and model interpretation of data
- Data and informed policy making
- Production = Expenditure = Income approaches; measuring GDP
- Price levels and inflation; real versus nominal
- Indexes
- Cross-country comparisons
- Yellow Labrador, Sunshine and lollipops

Acknowledgements

The figures and charts in this talk were created from:

- Google Maps
- Google Public Data
- Google Images